

Virtual Magnifying Glass

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About Virtual Magnifying Glass

Virtual Magnifying Glass is designed for visually-impaired and others who need to magnify a part of the screen. Unlike most similar programs it does not open a separate window for the magnification but instead puts a movable magnifying glass on screen.

This software is the multiplatform version of Virtual Magnifying Glass. Any feedback you can give us about this software will be greatly appreciated and can greatly enhance the quality of the software. You can give us feedback on our source-forge page:
<http://sourceforge.net/projects/magnifier>

Licence

Virtual Magnifying Glass is free software; you can redistribute it and/or modify it under the terms of the GNU General Public License version 2 as published by the Free Software Foundation.

Virtual Magnifying Glass is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

Please note that the General Public License version 2 does not permit incorporating Virtual Magnifying Glass into proprietary programs.

Installation Guide for Windows

Just follow the steps in the order they appear. If you need help installing, go to our website (<http://sourceforge.net/projects/magnifier>) and describe your problem.

- 1 - Download the .exe installation file to a known location.
- 2 - Execute the install software.
- 3 - Follow the instructions on the screen to install the software

Installation Guide for Linux and FreeBSD

Just follow the steps in the order they appear. If you need help installing, go to our website (<http://sourceforge.net/projects/magnifier>) and describe your problem.

- 1 - Download the magnifier tar.bz2 package to a known location
- 2 - Open a terminal session and change to root by issuing the "su" command
- 3 - Go to the directory where the package was downloaded
- 4 - Uncompress the file with the command: "tar -jxvf package_name.tar.bz2" e enter the directory which is created with "cd magnifier"
- 5 - Run the install script: ./install.sh
- 6 - If you wish to create a link on the desktop just type:

ln -s /usr/share/magnifier/magnifier \$HOME/Desktop/magnifier

In case you receive error messages, check if all the dependencies of the program are installed.

Note: There is also a uninstall script called uninstall.sh

Another Note: If your distribution doesn't use the standard directories for Install, please run magnifier once so it will create a .magnifier.conf file on your user directory (most times /home/username). Now edit that file and change the MyDirectory field to the correct location.

Important Note: In case you have problems executing the program, please run it from the command line so you can see any error messages.

Dependencies

This are the most important dependencies of the software:

- Gtk 2.6 or superior
- libgdk_pixbuf2

You can see a complete list of .so files needed by the software using the command:

```
ldd magnifier
```

In case you cannot run the program and it displays a message saying it cannot find one of this libraries (you must run the program from the command line to see error messages), you need to download and install a package for the necessary library:

On RPM-based distributions you can find packages here: <http://rpm.pbone.net/>

On Debian-based distributions you can find packages here:
<http://www.debian.org/distrib/packages>

Installation Guide for Mac OS X

Just follow the steps in the order they appear. If you need help installing, go to our website (<http://sourceforge.net/projects/magnifier>) and describe your problem.

- 1 - Download the .zip installation file to a known location.
- 2 - Uncompress the install .zip file.
- 3 - Drag the Magnifier application bundle to your "Applications" folder
- 4 - You can also drag the magnifier to the Dock to have a quick and comfortable way to start it.

Using the Glass

To start the software just double click it's shortcut on the Desktop.

Under Linux or FreeBSD you may also open a terminal and type: vmg <enter>

After it opens, Just move the cursor and the glass will follow it, enlarging the background with variable magnification from 1x to 16x.

<i>Key</i>	<i>Command</i>
[ENTER] or [ESC]	Hides the glass
Q	Closes the application
Up Arrow	Increases the lens height
Down Arrow	Decreases the lens height
Right Arrow	Increases the lens width
Left Arrow	Decreases the lens width
W	Moves the lens up
S	Moves the lens down
D	Moves the lens to the right
A	Moves the lens to the left
[PAGE UP]	Moves the lens up in big steps
[PAGE DOWN]	Moves the lens down in big steps
B	Activates and Deactivates the graphical border
[MOUSE WHELL UP]	Increase the magnification in small steps
[MOUSE WHELL DOWN]	Decrease the magnification in small steps
+	Increase the magnification in small steps
-	Decrease the magnification in small steps
Z	Decrease contrast
X	Increase contrast
C	Clear contrast
G	Show docked glass

Using the Dynamic Mode

The Dynamic Mode currently works under Windows 2000+. Unlike previous versions there is no longer the need to have DirectX installed or to have a special video card.

To activate the dynamic mode open the magnifier, right-click it's tray icon, select "Dynamic Mode" and then activate the glass. While using the dynamic mode the default magnifier keys won't work most of the time. We have, instead, special keys to move the glass while in Dynamic Mode. To exit the dynamic mode, right click the tray icon of the magnifier and quit it, or start a non-dynamic mode session.

The following keys are available to use the glass with the dynamic mode:

<i>Key</i>	<i>Command</i>
[Ctrl] + [Alt] + Up Arrow [Ctrl] + [Alt] + W [Ctrl] + [Alt] + 8	Moves the glass up
[Ctrl] + [Alt] + Down Arrow [Ctrl] + [Alt] + S [Ctrl] + [Alt] + 5	Moves the glass down
[Ctrl] + [Alt] + Right Arrow [Ctrl] + [Alt] + D [Ctrl] + [Alt] + 6	Moves the glass right
[Ctrl] + [Alt] + Left Arrow [Ctrl] + [Alt] + A [Ctrl] + [Alt] + 4	Moves the glass left

Using the Docked Mode

To use the glass in docked mode select: 'Show docked glass' from the systray menu.

To quickly launch the glass in docked mode use a hotkey. The default hotkey for docked mode is Ctrl + Alt + G (Linux users see the Linux shortcuts section to add a hotkey).

Another option to launch glass in docked mode is to press G while using the glass. This will lead to both modes being active at the same time.

To remove the borders of the docked glass window, right click inside the docked glass and disable show borders option.

To move the docked glass around drag the docked glass using the mouse

Image settings

There are some settings that can be applied to improve the enlarged image.

First of all anti-aliasing setting can be enabled from the menu on the systray. When the anti-aliasing setting is enabled, extra anti-aliasing can be enabled from the configurations dialog. Please note that this will lead to higher cpu usage.

Another setting is the contrast of the enlarged image.

Increase the contrast by pressing X button.

Decrease the contrast by pressing the Z button.

Clear the contrast by pressing the C button.

Linux shortcuts

To quickly launch the VMG application on your Linux system, you can add a custom shortcut to your desktop environment's settings.

For launching VMG in its default mode, use the following command: `vmg`

For launching VMG in docked mode, use the following command: `vmg -docked`

If you want to use keyboard to move the mouse around when in docked mode you should install `xdotool`.

For Debian/Ubuntu : `sudo apt install xdotool`

For Arch : `sudo pacman -S xdotool`

Then add custom shortcuts to these commands:

For moving right: `xdotool mousemove_relative 10 0`

For moving left: `xdotool mousemove_relative -- -10 0`

For moving up: `xdotool mousemove_relative -- 0 -10`

For moving down: `xdotool mousemove_relative 0 10`

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